project.org

TEACHER LESSON PLAN

How Vision Works

The Science of Seeing · Grades 5–8 · Science · 30 minutes

THE 30 MINUTES

5 min	10 min	12 min	3 min
HOOK	TEACH	ACTIVITY	CLOSING

OVERVIEW

Students explore eye anatomy and visual processing to understand why optical illusions exploit the limitations of human vision.

LEARNING OBJECTIVE

What students will understand

Students will understand the basic anatomy of the eye, explain how the brain processes visual information, and analyze why specific optical illusions exploit limitations in the visual system.

STANDARDS ALIGNMENT

NGSS MS-LS1-8: Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

MATERIALS NEEDED

- **Eye anatomy diagram** — included at end of this document
- **Blind spot test cards** — included at end of this document
- **Optical Illusions — Use the Mini Museum Perception Lab:** minimuseumproject.org/exhibitions/seeing-is-deceiving/illusions

Recommended illusions from Perception Lab: Hermann Grid (#20), Müller-Lyer (#12), Checker Shadow (#1), Lilac Chaser (#9 — afterimage), Simultaneous Contrast (#2), Ponzo Corridor (#10 — size constancy)

HOOK · 5 MINUTES

The Blind Spot Experiment

Setup: Distribute blind spot test cards. Instructions: (1) Close left eye, (2) Hold card at arm's length, focus on + with right eye, (3) Slowly bring card closer — the dot will disappear!

> *"The dot didn't actually disappear — your brain just can't see it. You have a REAL blind spot where the optic nerve connects. Your brain usually fills in this gap so you never notice. What does this tell us about how we see?"*

DIRECT INSTRUCTION · 10 MINUTES

Eye Anatomy & Visual Processing

Part 1: The Eye as a Camera (4 min) — Using eye diagram, explain path of light:

1. Cornea	Clear outer layer that bends (refracts) light
2. Pupil	Opening that controls how much light enters (like camera aperture)
3. Lens	Focuses light onto the back of the eye

4. Retina Light-sensitive layer with *rods* (light/dark, peripheral) and *cones* (color, center)

5. Optic nerve Carries electrical signals to brain (location of blind spot!)

Part 2: The Brain's Role (3 min)

> *"Your eyes don't 'see' anything. They just collect light and convert it to electrical signals. Your BRAIN creates the image. The retinal image is upside-down — your brain flips it. Your brain fills in gaps and makes predictions. Processing happens in the visual cortex at the back of your brain."*

Part 3: Why This Matters (3 min) — Illusions exploit these processes:

- **Eye-based:** Afterimages (tire out cones), peripheral motion (rods vs. cones)
- **Brain-based:** Size illusions (context clues), ambiguous figures (brain picks interpretation)

ACTIVITY · 12 MINUTES

Analyzing Illusions Scientifically

Present 3–4 illusions; students identify WHERE in visual system each occurs:

01

Illusion 1: Color Afterimage

Stare at colored image 30 sec, look at white paper.

What: See opposite colors

Why: Cone cells fatigue; opposite cones fire stronger

Location: Retina (eye-based)

02

Illusion 2: Hermann Grid

White grid on black — gray dots at intersections.

What: See dots that aren't there

Why: Lateral inhibition — edge cells suppress neighbors

Location: Retina + early processing

03

Illusion 3: Müller-Lyer

Equal lines look different due to arrow endings.

What: One line looks longer

Why: Brain interprets arrows as depth cues

Location: Visual cortex (brain-based)

04

Illusion 4: Stereoscope

Two flat images become 3D.

What: Depth from flat images

Why: Each eye sees different image; brain combines

Location: Visual cortex

CLOSING · 3 MINUTES

Connection to Critical Thinking

> *"Your visual system is sophisticated — but it has real limitations. Every illusion teaches us something about how we see. If your brain can be fooled by simple lines and colors, what else might it misinterpret? This is why scientists don't just trust what they see — they measure, test, and verify."*

MINI MUSEUM CONNECTION

Option 1 — Visit: The 1904 Monarch Stereoscope demonstrates binocular vision. Each Keystone view card has two slightly different photographs — one for each eye. Your visual cortex fuses them into a single 3D scene.

Option 2 — Virtual: Explore the exhibition online:

minimuseumproject.org/exhibitions/seeing-is-deceiving

ASSESSMENT

Formative: Check that students can identify whether illusions are eye-based or brain-based and explain the mechanism.

Exit ticket: "Choose one illusion. Explain where in the visual system it occurs and why it works, using at least two vocabulary terms."

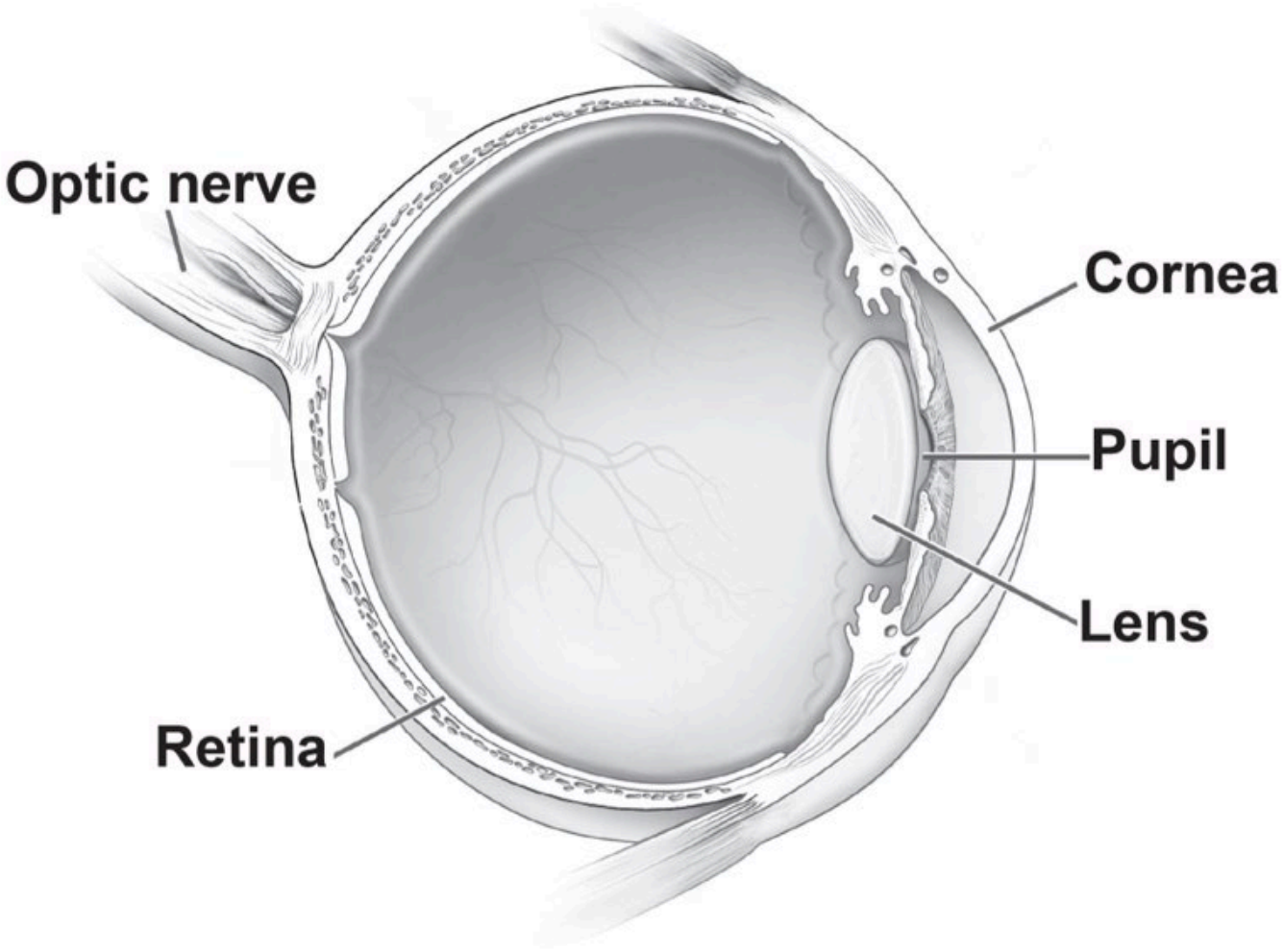
DIFFERENTIATED STUDENT HANDOUTS

- **Green:** Labeled diagram, guided analysis, word bank
- **Yellow:** Label diagram independently, analyze 2–3 illusions
- **Red:** Research specific illusions, explain neuroscience mechanisms

SHOW YOUR CLASS

Eye anatomy diagram

For Part 1 of the direct instruction: project this page, or print it, and follow the path of light from the cornea to the optic nerve.



TEACHER INSTRUCTIONS FOR THE SCIENCE LESSON ON HOW VISION WORKS

Blind Spot Test Cards

WHAT IS THE BLIND SPOT?

Every human eye has a **blind spot** — a small area where there are no photoreceptors (rods or cones) because this is where the optic nerve connects to the retina. Your brain usually "fills in" this gap using information from the surrounding area, so you never notice it in daily life.

HOW TO USE THE CARDS

1. **Photocopy page 5** and cut out the cards along the dashed lines (2 cards per page)
2. Give one card to each student (or pair of students)
3. Have students **close their LEFT eye**
4. Hold the card at **arm's length**
5. Focus on the **+** with the RIGHT eye only
6. **Slowly bring the card closer** to the face
7. At approximately 10–12 inches, the dot will **disappear!**

DIAGRAM



THE SCIENCE BEHIND IT

When the dot falls on the blind spot (where the optic nerve exits the eye), there are no photoreceptors to detect it. The brain automatically fills in the missing information using the surrounding white color — so the dot seems to vanish! This demonstrates that we don't see the world directly; our brain *constructs* our visual experience.

DISCUSSION QUESTIONS

- Why don't we notice our blind spot in everyday life?
- What does this tell us about how our brain processes visual information?
- How does this connect to why optical illusions can fool us?

BLIND SPOT TEST CARDS

Photocopy this page · Cut along dashed lines · 2 cards per page

Cut along dashed line



— cut here —

Cut along dashed line

